JOSHUA COWELL

Lynchburg, VA | joshuacowell2005@gmail.com | cowelljoshua.github.io

EDUCATION

Bachelor of Science in Mechanical Engineering Expected May 2027
Liberty University, Lynchburg, VA GPA: 3.82

Relevant Coursework: Fluid Dynamics, Heat Transfer, Thermodynamics I & II, Differential Equations, Dynamics, Materials Engineering, Statics, Calculus I–III

Activities: American Nuclear Society (ANS) – Founding Member (Nov 2025)

PROFESSIONAL EXPERIENCE

Thermal Hydraulics Engineering Intern May 2025 – Dec 2025
Framatome Nuclear

- Developed a predictive wear model in Python for safety-critical reactor components, achieving 75% improvement in accuracy over legacy methods
- Engineered a Python algorithm processing 6-dimensional datasets to calculate safe operating limits, reducing analysis time from 800+ hours to minutes
- Performed system-level thermal-hydraulic and flow-induced vibration (FIV) analysis to support component qualification
- Authored technical documentation and presented findings to engineering leadership; awarded $2,500 for presentation excellence

Chief Systems Engineer and Team Lead June 2025 – Present
Liberty Rocketry (9th of 156 international teams, 2024)

- Lead a cross-functional team of 60+ members in the design, integration, and verification of high-powered rocket systems
- Direct systems engineering decisions across propulsion, avionics, recovery, and aerodynamics subsystems
- Manage requirements definition, interface control, and configuration management throughout the development lifecycle
- Oversee a $16,000+ project budget, vendor coordination, and schedule milestones for competition campaigns
- Perform verification and validation (V&V) using SolidWorks, ANSYS, and MATLAB for subsystem models

Lead Undergraduate Research Assistant Sep 2024 – Present
Liberty University

- Lead a team of 5 researchers conducting split-Hopkinson pressure bar (SHPB) testing for dynamic material characterization
- Design test procedures and fixtures for high-strain-rate experiments; analyze results using electron microscopy and high-speed imaging

Senior Structural Engineer Sep 2023 – June 2025
Liberty Rocketry

- Conducted finite element analysis (FEA) and computational fluid dynamics (CFD) simulations to verify structural integrity and aerodynamic performance
- Authored technical reports documenting analysis methodology, results, and compliance with competition requirements

Camp Counselor May 2024 – Aug 2024
Woodlands Camp, GA

- Provided 24-hour supervision during weeklong outdoor expeditions with 12 youth (ages 8–18), instructing survival skills, rock climbing, whitewater rafting, and cave camping

TECHNICAL SKILLS

Analysis & CAE:	ANSYS Fluent, ANSYS Mechanical, AFT Fathom, OpenRocket, RocketPy, FEA, CFD
CAD & Design:	SolidWorks, Onshape, Technical Drawing, 3D Printing, CNC Machining
Programming:	Python (NumPy, Xarray, Pandas), MATLAB, Data Analysis, Automation
Laboratory:	Scanning Electron Microscope (SEM), High-Speed Imaging, Material Testing

PROJECTS

Predictive Wear Model – Nuclear Component Lifecycle Analysis

- Developed a Python-based predictive model for safety-critical components, improving failure prediction accuracy by 75%

Automated Operating Limit Analysis Tool

- Engineered a Python automation framework reducing 800+ hours of manual engineering calculations to minutes

Heat Exchanger Thermal-Hydraulic Optimization

- Performed CFD analysis using ANSYS Fluent to optimize heat exchanger thermal performance and flow characteristics

Unmanned Aerial Vehicle (UAV) Design & Analysis

- Designed a modular UAV airframe with rapid-prototyping capability; validated aerodynamics through CFD simulation

3D Printed Model Rocket

- Engineered a fully 3D printed rocket achieving Mach 0.5+ and 3,000 ft altitude; performed CFD analysis for aerodynamic optimization

High Strain Rate Bone Research

- Investigating the dynamic loading behavior of human bone using Split Hopkinson Pressure Bar for conference presentation